

Questions with Answer Keys

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Q1: 24 Feb (Shift 2) - Numerical

The volume occupied by 4.75 g of acetylene gas at 50°C and 740 mmHg pressure is _____ L.

(Rounded off to the nearest integer) (Given $R = 0.0826 \text{ L atm K}^{-1} \text{ mol}^{-1}$)

Q2: 25 Feb (Shift 1) - Numerical

A car tyre is filled with nitrogen gas at 35 psi at 27°C . It will burst if pressure exceeds 40 psi. The temperature in $^{\circ}\text{C}$ at which the car tyre will burst is _____ (Rounded-off to the nearest integer)

Q3: 26 Feb (Shift 1) - Numerical

3.12 g of oxygen is adsorbed on 1.2 g of platinum metal. The value of oxygen adsorbed per gram of the adsorbent at 1 atm and 300 K in L is _____ $\left[R = 0.0821 \text{ L atm K}^{-1} \text{ mol}^{-1}\right]$

Q4: 26 Feb (Shift 1) - Numerical

A certain gas obeys $P(V_m - b) = RT$. The value of $\left(\frac{\partial Z}{\partial P}\right)_T$ is $\frac{xb}{RT}$. The value of x is _____

Answer Key**Q1 (5)****Q2 (70)****Q3 (2)****Q4 (1)**

Q1 (5)

$$T = 50^{\circ}\text{C} = 323.15 \text{ K}$$

$$P = 740 \text{ mm of Hg} = \frac{740}{760} \text{ atm}$$

$$V = ?$$

$$\text{moles (n)} = \frac{4.75}{26}$$

$$V = \frac{4.75}{26} \times \frac{0.0821 \times 323.15}{740} \times 760$$

$$V = 4.97 \simeq 5 \text{ Lit}$$

Q2 (70)

$$\frac{P_1}{T_1} = \frac{P_2}{T_2}$$

$$\frac{35}{300} = \frac{40}{T_2}$$

$$T_2 = \frac{40 \times 300}{35}$$

$$= 342.86 \text{ K}$$

$$= 69.85^{\circ}\text{C} = 70^{\circ}\text{C}$$

Q3 (2)

$$\text{Moles of O}_2 = \frac{3.12}{32} = 0.0975$$

$$\text{volume of O}_2 = \frac{nRT}{p} = \frac{0.0975 \times 0.082 \times 300}{1}$$

$$= 2.3985 \text{ L} \simeq 2.4 \text{ L}$$

$$\text{volume of O}_2 \text{ absorbed per gm of pt} = \frac{2.4}{1.2} = 2$$

Q4 (1)

$$P(v - b) = RT$$

$$PV - Pb = RT$$

$$\frac{PV}{RT} - \frac{Pb}{RT} = 1$$

$$Z = 1 + \frac{PV}{RT}$$

$$\frac{dz}{dp} = 0 + \frac{b}{RT}$$

Hints and Solutions

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$$\Rightarrow \frac{b}{RT} = \frac{xb}{RT}$$

$$x = 1$$